Dr Anton Baykalov

Curriculum Vitae

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Education

2017–2021 **PhD in Mathematics**, *The University of Auckland, New Zealand*.

Supervisors Professor Eamonn O'Brien & Associate Professor Jianbei An

Thesis Intersection of conjugate solvable subgroups in finite classical groups

Description In this thesis I solve Problem 17.41 from the Kourovka notebook for the case of an almost simple group G with the socle isomorphic to $L_n(q)$, $U_n(q)$ or $PSp_n(q)'$.

2014–2016 **Master of Mathematics and Computer Science**, *Novosibirsk State University, Russia*.

2010–2014 **Bachelor of Mathematics with thesis**, *Novosibirsk State University, Russia*.

Publications

1 Anton A. Baykalov, Alice Devillers and Cheryl E. Praeger. *Proper partial linear spaces affording imprimitive rank 3 automorphism groups*, Journal of Algebra 688 (2026), 454-526. https://doi.org/10.1016/j.jalgebra.2025.09.020

2 Anton A. Baykalov. *Base sizes for finite unitary and symplectic groups with solvable stabilisers*, International Journal of Algebra and Computation 35 (2025), no. 6, 823-908. https://doi.org/10.1142/S0218196725500237

3 Anton A. Baykalov. *Base sizes for finite linear groups with solvable stabilisers*, Journal of Group Theory 28 (2025), no. 5, pp. 1003-1077. https://doi.org/10.1515/jgth-2023-0125

4 Anton A. Baykalov. *On algebraic normalisers of maximal tori in simple groups of Lie type*, Journal of Group Theory, vol. 27, no. 6, 2024, pp. 1357-1383.) https://doi.org/10.1515/jgth-2023-0070

5 Anton A. Baykalov, Alice Devillers and Cheryl E. Praeger. *Rank three innately transitive permutation groups and related 2-transitive groups*, **Innov. Incidence Geom.** 20(2023), no.2-3, 135–175. https://doi.org/10.2140/iig.2023.20.135

6 Anton A. Baykalov. *Intersection of conjugate solvable subgroups in finite classical groups.* **PhD Thesis** (2021), The University of Auckland. https://arxiv.org/abs/1703.00124

7 Anton Baikalov. *Intersection of conjugate solvable subgroups in symmetric groups.* **Algebra Logic** 56, 87–97 (2017). https://doi.org/10.1007/s10469-017-9431-z

Work experience

Jan 2025 – present time	Postdoctoral researcher, School of Mathematical and Statistical Sciences, University of Galway.
June 2023 – Dec 2024	Research Associate, Department of Mathematics and Statistics, The University of Western Australia.
Jan – June, 2023	Temporary Lecturer, Department of Mathematics, The University of Auckland.
Jan – June, 2022	Research Assistant, The University of Auckland. Preparing MAGMA packages for computing representatives of conjugacy classes of finite simple groups of Lie type G_2 and F_4 together with Professor Eamonn O'Brien.

Teaching experience

2023	Teaching the course MATHS 254 (Fundamental Concepts of Mathematics, for undergraduate students, class of about 60 students) at the University of Auckland
2021	Teaching the course MATHS 326 (Combinatorics, for undergraduate students, class of about 30 students) at the University of Auckland
2019	Tutoring for the course MATHS 120 (an introduction to linear algebra) at the University of Auckland
2016–2017	Tutoring for the Algebra course (linear algebra and introduction to groups, rings and other algebraic structures) at Novosibirsk State University

Academic service roles

Sep 2025 – present time	A member of Research and Graduate Studies Committee, School of Mathematical and Statistical Sciences, University of Galway.
Jan 2025 – present time	Convenor of a reading group "Machine Learning for Mathematics", School of Mathematical and Statistical Sciences, University of Galway.
July 2023 – Dec 2024	Convenor for the Groups and Combinatorics seminar, Department of Mathematics and Statistics, The University of Western Australia.

Outreach

2010-2016	Teaching Olympiad mathematics and related topics to middle-school and high-school students on various summer schools and in school math clubs in Novosibirsk including four semesters (2014–2016) of teaching twice a week in Novosibirsk math club "Sov'onok."

Languages

Russian	**Mother tongue**
English	**Fluent**